



Student Name: Ridhima Gulati
Branch: AIT-CSE (Cyber Security)
Semester: 4
Subject Name: Database Management System

UID: 24BCY70132
Section/Group: 24AIT_KRG-G2
Date of Performance: 27/02/2026
Subject Code: 24CSH-298

WORKSHEET 6

AIM: To understand the concept and working of cursors in PL/SQL for row-by-row data processing, and to analyze how implicit cursors, explicit cursors, and cursor attributes are used to implement business logic on multiple rows in a database table.

S/W Requirement: • Database Management System: PostgreSQL / Oracle Database Express Edition
• Database Administration Tool: pgAdmin

OBJECTIVES:

To implement and analyze the use of:

- Implicit cursors
- Explicit cursors
- Cursor attributes

for processing multiple rows from a database table and applying business logic effectively.

PROBLEM STATEMENT:

In real-world enterprise applications, database queries often return multiple rows that need to be processed individually to apply specific business rules.

PRACTICAL/EXPERIMENT STEPS:

Design and execute PL/SQL programs that demonstrate:

- The use of implicit cursors for single-row DML operations
- The use of explicit cursors to fetch and process multiple records
- The application of cursor attributes to control program execution.

1. PROBLEM STATEMENT – Implicit Cursor

Write a PL/SQL program to update salary and display the number of rows affected using an implicit cursor.

PROGRAM:

```
BEGIN
    UPDATE Employee_data
    SET salary = salary + 1000
    WHERE emp_id = 101;

    IF SQL%FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || SQL%ROWCOUNT);
    ELSE
        DBMS_OUTPUT.PUT_LINE('No rows updated');
    END IF;
END;
```

OUTPUT:

Rows Updated: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.011

2. PROBLEM STATEMENT – Explicit Cursor

Fetch and display employee records using an explicit cursor.

PROGRAM:

```
25 > DECLARE
26 >     CURSOR emp_cursor IS
27 >         SELECT emp_id, emp_name, salary FROM Employee_data;
28 >         v_id employee.emp_id%TYPE;
29 >         v_name employee.emp_name%TYPE;
30 >         v_salary employee.salary%TYPE;
31 > BEGIN
32 >     OPEN emp_cursor;
33 >
34 >     LOOP
35 >         FETCH emp_cursor INTO v_id, v_name, v_salary;
36 >         EXIT WHEN emp_cursor%NOTFOUND;
37 >
38 >         DBMS_OUTPUT.PUT_LINE(v_id || ' ' || v_name || ' ' || v_salary);
39 >     END LOOP;
40 >
41 >     CLOSE emp_cursor;
42 > END;
```

OUTPUT:

```
101 John 4000
102 Alice 4000
103 Bob 3500
104 Emma 4500
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.107

3. PROBLEM STATEMENT –Cursor Attributes

Demonstrate the use of cursor attributes such as %FOUND, %NOTFOUND, and %ROWCOUNT.

PROGRAM:

```
44 DECLARE
45     CURSOR emp_cursor IS
46         SELECT emp_id FROM Employee_data;
47
48     v_id employee.emp_id%TYPE;
49 BEGIN
50     OPEN emp_cursor;
51
52     LOOP
53         FETCH emp_cursor INTO v_id;
54
55         IF emp_cursor%FOUND THEN
56             DBMS_OUTPUT.PUT_LINE('Employee ID: ' || v_id);
57         END IF;
58
59         EXIT WHEN emp_cursor%NOTFOUND;
60     END LOOP;
61
62     DBMS_OUTPUT.PUT_LINE('Total Rows Fetched: ' || emp_cursor%ROWCOUNT);
63
64     CLOSE emp_cursor;
65 END;
```

OUTPUT:

```
Employee ID: 101
Employee ID: 102
Employee ID: 103
Employee ID: 104
Total Rows Fetched: 4
```

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006



LEARNING OUTCOMES:

1. Understood the role of cursors in PL/SQL for handling multi-row query results
2. Learned the difference between implicit and explicit cursors
3. Used cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN
4. Developed PL/SQL programs to process records row by row
5. Applied cursor-based logic in practical database scenarios.

CONCLUSION:

This experiment demonstrated the use of implicit and explicit cursors in Oracle PL/SQL.